

Projection-Based Handling of Contact Constraints

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CCS CONCEPTS

• **Mathematics of computing** → **Quadratic programming**; •
Computing methodologies → *Physical simulation*.

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1 INTRODUCTION

Simulating contact of deformable objects has many different challenging aspects. In general, contact simulation is a highly nonlinear process as contact constraints ensuring no penetration occurs depend on the deformed state. In the simplest case contact can be handled using penalty methods, i.e. inserting stiff springs between the deformables. This has the advantage, that the system remains positive definite and can be solved using iterative solvers, such as the preconditioned conjugate gradient (PCG) method. However, besides not providing accurate results in terms of constraint satisfaction, penalty-methods also worsen the condition of the system matrix. Alternatively, non-linear optimization approaches can be used, such as the barrier-method [Li et al. 2020] or sequential quadratic programming methods. In both cases indefinite systems need to be solved, for which generally direct solvers are used (e.g. [Cheshmi et al. 2020]). For very high degrees of freedom iterative solvers such as the PCG method are however favorable due to lower run times and lower memory usage.

In this work, we focus on handling linearized contact, which can be formulated as mixed linear complementary problem or equivalently as quadratic program [Verschoor and Jalba 2019], which is solved using the Lagrange multiplier method. We propose using projection-based methods for handling constraints arising in contact simulation combined with the PCG method. We are able to provide high accuracy in terms of constraint satisfaction by using the Lagrange multiplier approach, while being able to use a time and memory efficient iterative solver for large simulations.

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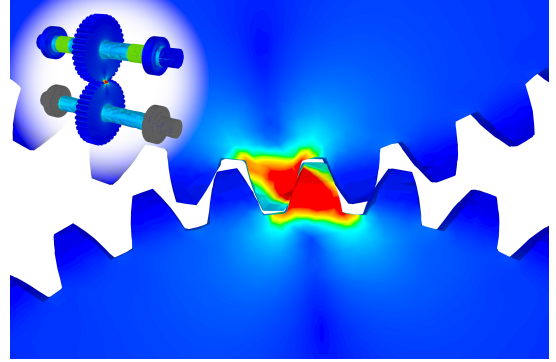


Figure 1: Evaluation scenario: A torque is applied to both ends of the top gear (blue nodes), which is only able to rotate around its axis (enforced by a boundary condition on the green nodes). The bottom gear is blocked by fixation of the grey nodes. This results in contact forces occurring. Red color indicates high stress.

2 BACKGROUND

Using the finite element method, the discretized state of an object can be modeled as linear system of equations $\mathbf{K}\mathbf{u}^* = \mathbf{f}$, where $\mathbf{K} \in \mathbb{R}^{3N \times 3N}$ is the stiffness matrix and $\mathbf{u} \in \mathbb{R}^{3N}$ is the displacement vector of the discretized object with N nodes. The stacked positions $\mathbf{p}$ of all nodes in the state of equilibrium for acting forces $\mathbf{f} \in \mathbb{R}^{3N}$ are given by $\mathbf{p} = \mathbf{p}_0 + \mathbf{u}^*$, with $\mathbf{p}_0$ being the positions in the rest state. Equivalently, $\mathbf{u}^*$ can be computed as solution of a quadratic minimization problem. Contact constraints ensuring no penetration occurs can be formulated as inequality constraints acting on this energy minimization problem. Here, we focus on the linearized distance function $g(\mathbf{u}) = \mathbf{C}\mathbf{u} - g_0$ describing the distance between primitive pairs (node-triangle and edge-edge), which potentially can come into contact during a simulation step. Each row of $\mathbf{g}$ describes the linearized distance function for a single contact pair. This results in a quadratic program (QP):

$$\begin{aligned} \min_{\mathbf{u}} \quad & \mathbf{u}^T \mathbf{K} \mathbf{u} - \mathbf{u}^T \mathbf{f} \\ \text{s.t.} \quad & \mathbf{C}_{\text{ineq}} \mathbf{u} - \mathbf{g}_0 \geq 0. \end{aligned} \quad (1)$$

To solve this QP we chose a primal active-set method, as we expect the number of active constraints at the optimal solution to be low even though the number of potential contacts can be very high. In each active-set iteration k an equality constrained QP needs to be

solved to find the step direction $\mathbf{d}_k$:

$$\begin{aligned} \min_{\mathbf{d}_k} \mathbf{d}_k^T \mathbf{K} \mathbf{d}_k - \mathbf{d}_k^T (\mathbf{f} - \mathbf{K} \mathbf{u}_k) \\ \text{s.t. } \mathbf{C}_{\mathcal{A}_k} \mathbf{d}_k = \mathbf{0}, \end{aligned} \quad (2)$$

where $\mathbf{C}_{\mathcal{A}_k}$ refers to the submatrix of $\mathbf{C}_{\text{ineq}}$ containing the rows which are currently considered to be active. In each active-set iteration a step is taken in the computed direction: $\mathbf{u}_{k+1} = \mathbf{u}_k + \alpha \mathbf{d}_k$, with the step length α [Nocedal and Wright 2006].

3 APPROACH

In this work, we extend the modified preconditioned conjugate gradient (MPCG) method, a projection-based approach for handling node constraints, to handle constraints involving multiple nodes to solve the equality constrained system in Equation (2). The MPCG method was introduced by Baraff and Witkin [1998] (see also [Ascher and Boxerman 2003]) and adapted by Tamstorf et al. [2015] for better preconditioning performance. The idea is to (implicitly) transform the constrained minimization problem into an unconstrained problem, solvable using the conjugate gradient method. For this, an orthogonal projection matrix (filter matrix) is defined, which performs a projection onto the feasible space. For node constraints this projection matrix is a block diagonal matrix which can be directly derived from the constraint definition. For constraints involving multiple nodes the computation however is not straight forward. Firstly, one needs to note, that the feasible space is given by the shifted null space of the active constraint matrix $\mathbf{C}_{\mathcal{A}_k}$. As long as it is ensured that the step direction $\mathbf{d}_k$ is chosen such that it lies in the null space of $\mathbf{C}_{\mathcal{A}_k}$, the solution $\mathbf{u}_{k+1}$ remains primal feasible, i.e. fulfills the contact constraints.

The orthogonal projection matrix onto the null space of a matrix $\mathbf{C} \in \mathbb{R}^{m \times 3N}$ can be computed as $\mathbf{P}_C = \mathbf{Z}_C \mathbf{Z}_C^T$, with the null space matrix $\mathbf{Z}_C \in \mathbb{R}^{3N \times 3N-m}$, where the column vectors form an orthonormal basis of the null space. This projection matrix can be used inside the conjugate gradient method as filter matrix equivalently to the MPCG method.

As preconditioning is challenging in this setting, the projection matrix can be used to create the prefiltered system, as proposed by Tamstorf et al. [2015]. Following this, Equation (2) can be reformulated as $\tilde{\mathbf{K}} \tilde{\mathbf{d}} = \tilde{\mathbf{f}}$, where

$$\tilde{\mathbf{K}} = \mathbf{P}_C \mathbf{K} \mathbf{P}_C + \mathbf{I} - \mathbf{P}_C, \quad (3)$$

with the $(3N \times 3N)$ identity matrix $\mathbf{I}$. $\tilde{\mathbf{K}}$ is a symmetric positive definite matrix, where the constrained degrees of freedom are removed by applying the projection, while keeping the dimensions and the block structure of the matrix. The PCG method can be used to solve this resulting system to obtain a feasible step direction. We extend the prefiltered approach by adding a final post-filtering step:

$$\mathbf{d}_k = \mathbf{P}_C \tilde{\mathbf{d}}, \quad (4)$$

instead of directly setting $\mathbf{d}_k = \tilde{\mathbf{d}}$, as it is possible for node constraints. This step is crucial for more complex constraints, due to numerical errors outside the null space are accumulated during the minimization process in the PCG method. These errors are completely removed by the post-filtering step, providing a solution within the requested accuracy.

Solver	MPCG	Prefiltering and PCG
Solve time	6.8 s	11.2 s
Residual	$5.1 \cdot 10^{-4}$	$2.8 \cdot 10^{-4}$
Max. constraint violation	$1.9 \cdot 10^{-19}$	$8.5 \cdot 10^{-20}$

Table 1: Results for solving the QP using the extended MPCG solver and the prefiltered approach with a PCG solver computed on a Windows system with a Intel(R) i9-13900K CPU and a NVIDIA(R) GeForce RTX 4090 graphics card, for a target accuracy of 10^{-3} , which required 4 active-set iterations.

4 RESULTS

To achieve high performance, our solver exploits parallelizability of the PCG method. We solved the simulation scenario of two gears coming into contact in the scenario shown in Figure 1, with a total of 2.2 million degrees of freedom and about 1 thousand potential contacts, of which three are active at the computed optimal solution. The results and runtimes are shown in Table 1. The main advantage of using a projection-based approach in combination with the PCG method is, that the constraints can be enforced with very high accuracy, independantly of the termination criteria chosen for the PCG method. This enables more freedom and control over the runtime without sacrificing constraint accuracy. Especially, in structural simulation it is common to use relative thresholds as termination criteria. The presented approach allows larger absolute thresholds (resulting from relative termination criteria), while enforcing constraints with high accuracy. A further exploitation of this property could be to first solve the QP approximately, to acquire a close to optimal solution and its corresponding active-set, which can then be used to warm-start a second run of the solver with the requested termination criteria, potentially decreasing the overall runtime.

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